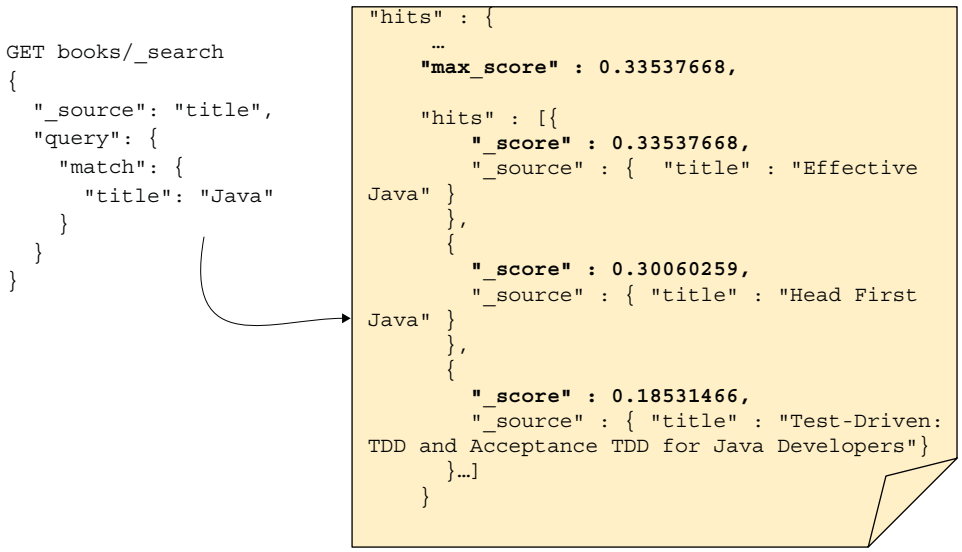



**Figure 3.19 Relevant results for *Java* in a title search**

The first result shows a higher relevancy score (`0.33537668`) than the second and third results. All the titles include the word *Java*, so to calculate the relevancy score, Elasticsearch uses the field-length norm algorithm: the search word in the first title (*Effective Java*, containing two words) is more relevant than the search word in the second title (*Head First Java*, consisting of three words). The `max_score` is the highest of all the available scores, usually the score of the first document.

Relevancy scores are generated based on the similarity algorithms employed. In this example, Elasticsearch by default applied the BM25 algorithm to determine the scores. This algorithm depends on term frequency, inverse document frequency, and field length, but Elasticsearch allows much more flexibility: it provides a handful of algorithms in addition to BM25. These algorithms are packaged in a module called `similarity`, which ranks matching documents. Let's look at some of these similarity algorithms.

### 3.4.2 Relevancy (similarity) algorithms

Elasticsearch employs several relevance algorithms, the default being BM25. BM25 and TF-IDF are both relevance algorithms used in Elasticsearch for scoring and ranking documents. They differ in how they calculate term weights and document scores.

- *TF-IDF (Term Frequency-Inverse Document Frequency).* The TF-IDF algorithm is a traditional weighting scheme that assigns a weight to a term in a document based on its term frequency (TF) and inverse document frequency (IDF). TF is the number of times a term appears in a document, while IDF is a measure of how common or rare a term is across the entire set of documents. The terms that